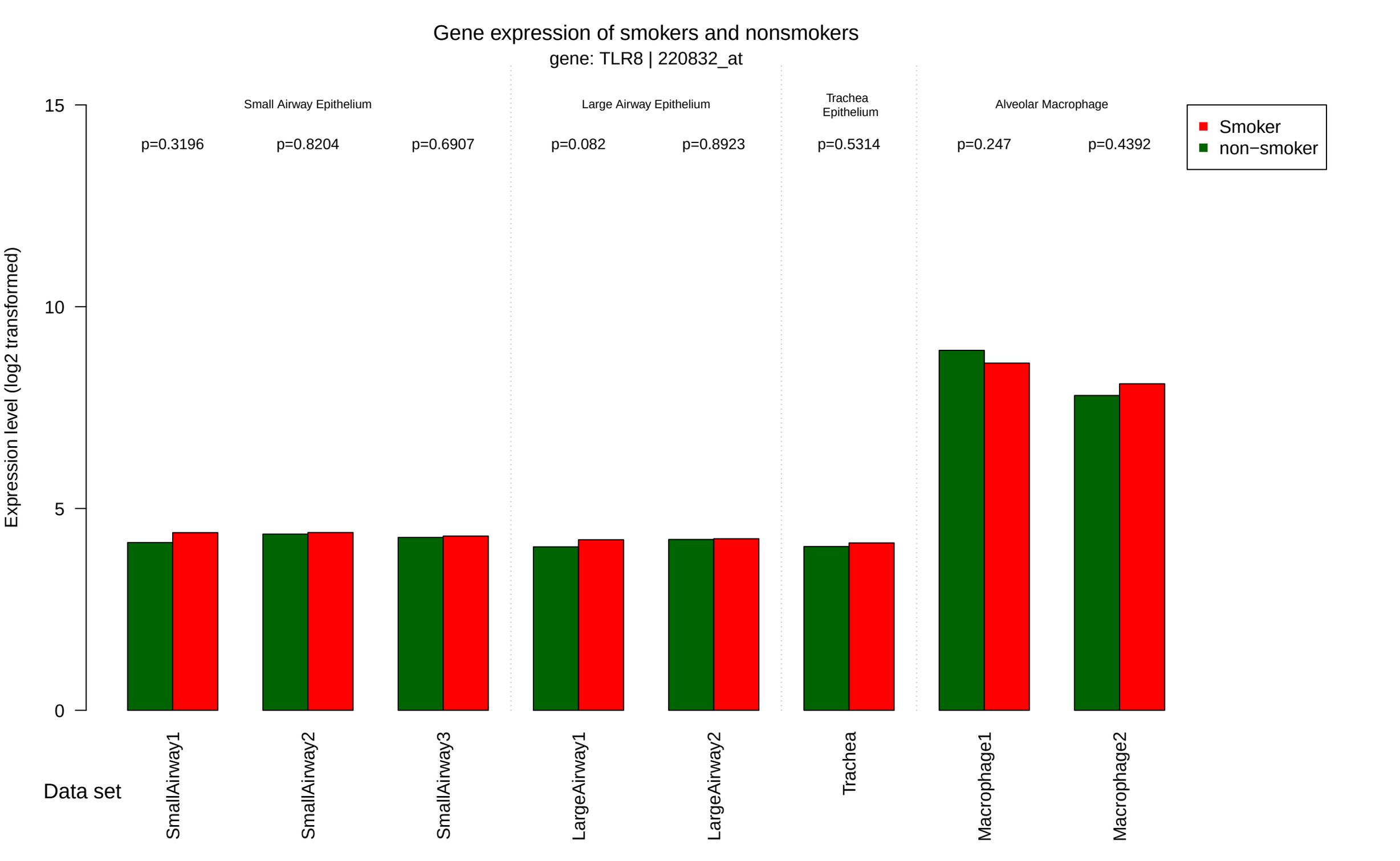
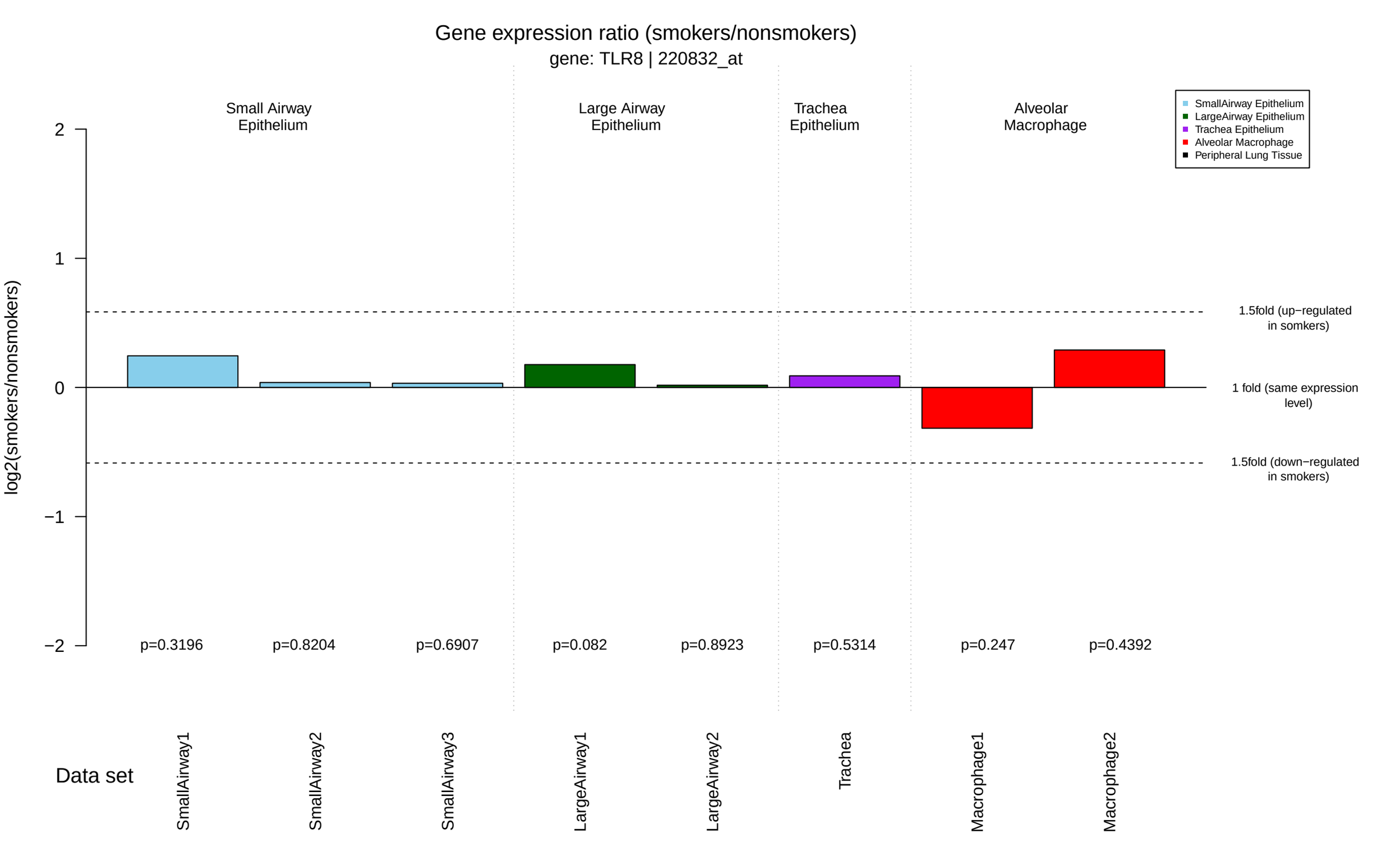
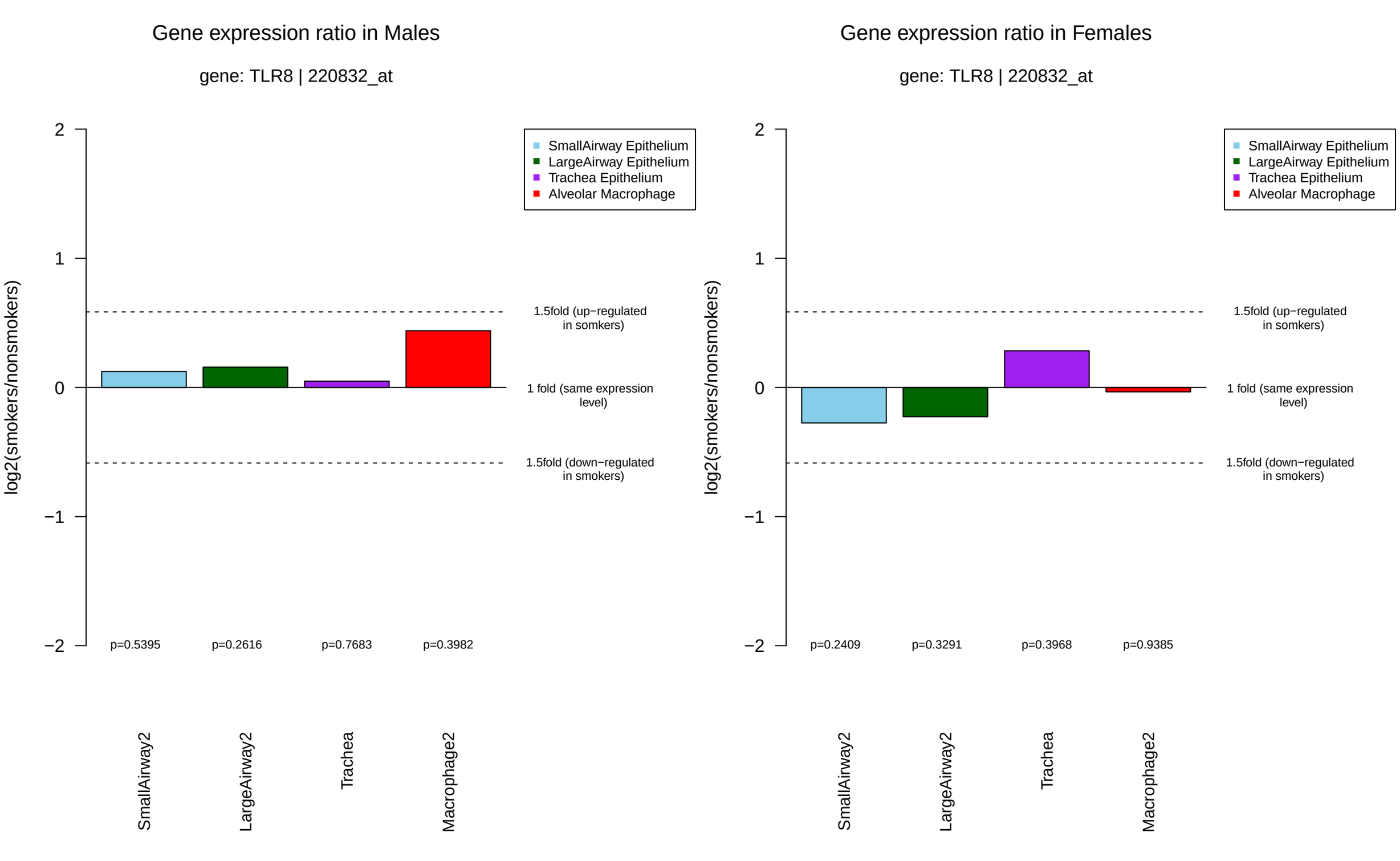
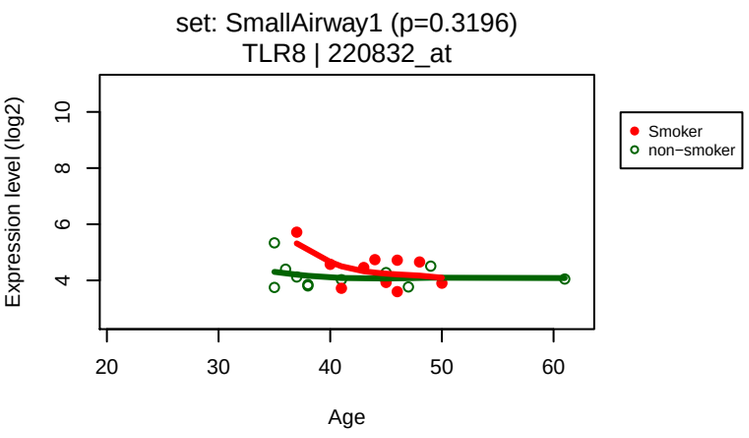
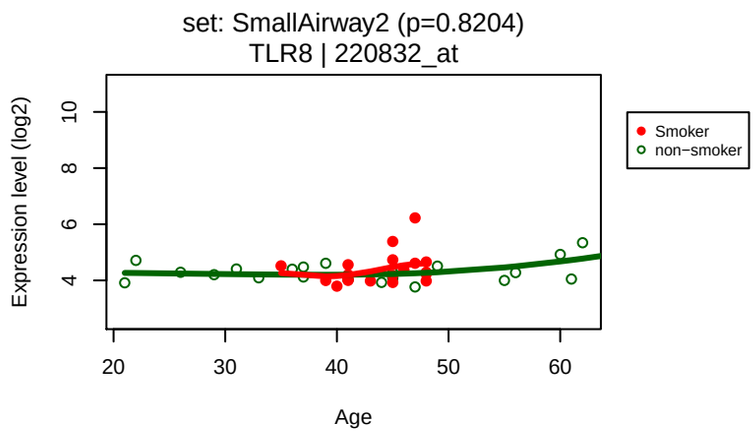
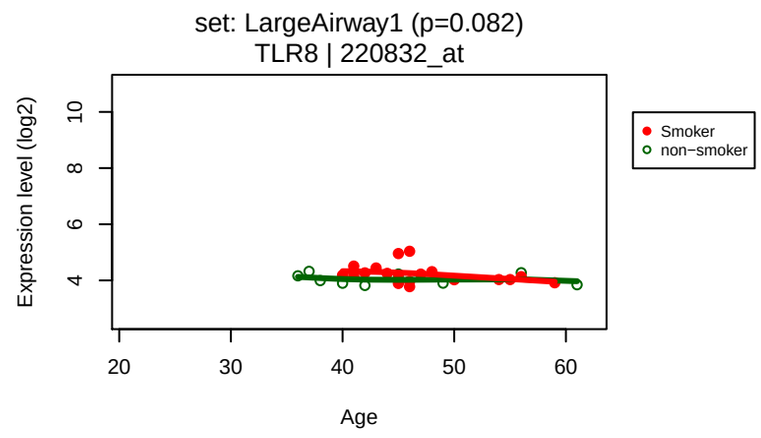

Gene expression of smokers and nonsmokers
gene: TLR8 | 220832_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3196
p=0.8204
p=0.6907
p=0.082
p=0.8923
p=0.5314
p=0.247
p=0.4392
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

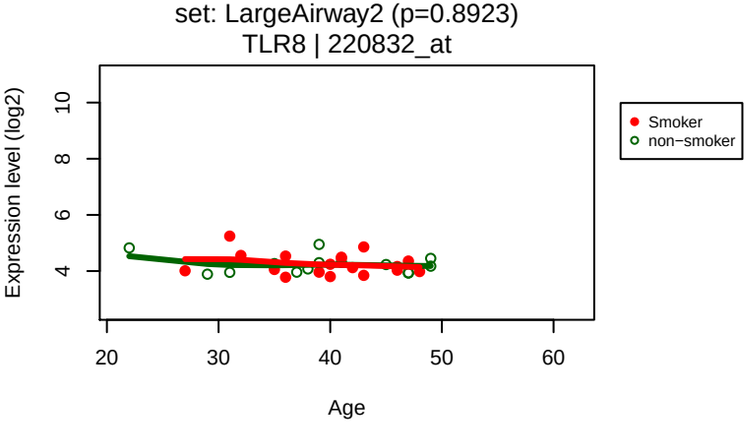

Gene expression ratio (smokers/nonsmokers)
gene: TLR8 | 220832_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3196
p=0.8204
p=0.6907
p=0.082
p=0.8923
p=0.5314
p=0.247
p=0.4392
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

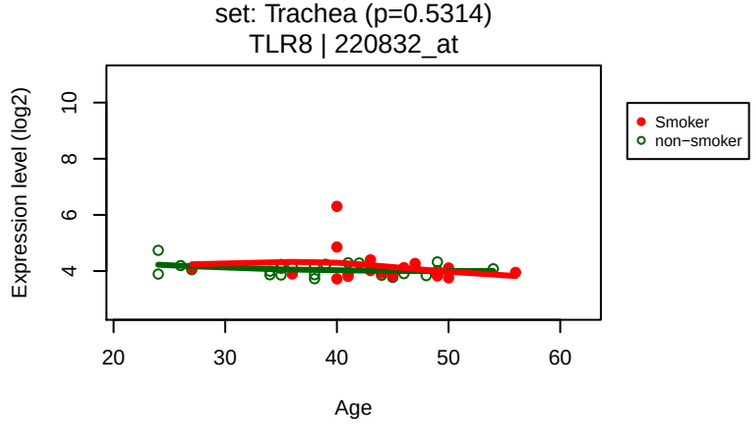

Gene expression ratio in Males
gene: TLR8 | 220832_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5395
p=0.2616
p=0.7683
p=0.3982
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR8 | 220832_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2409
p=0.3291
p=0.3968
p=0.9385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

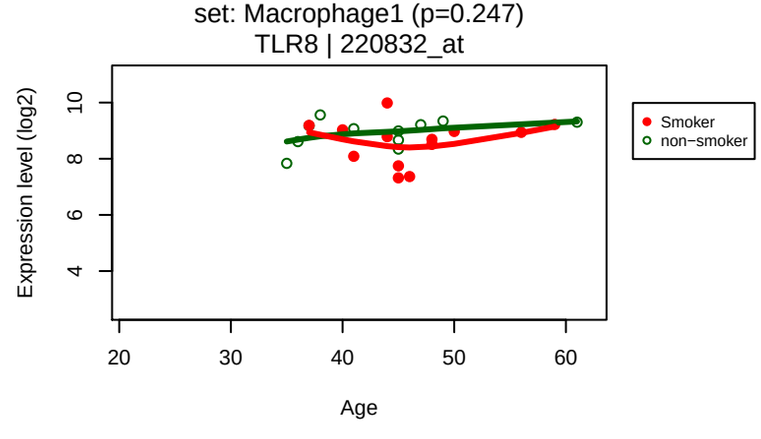

set: SmallAirway1 (p=0.3196)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker

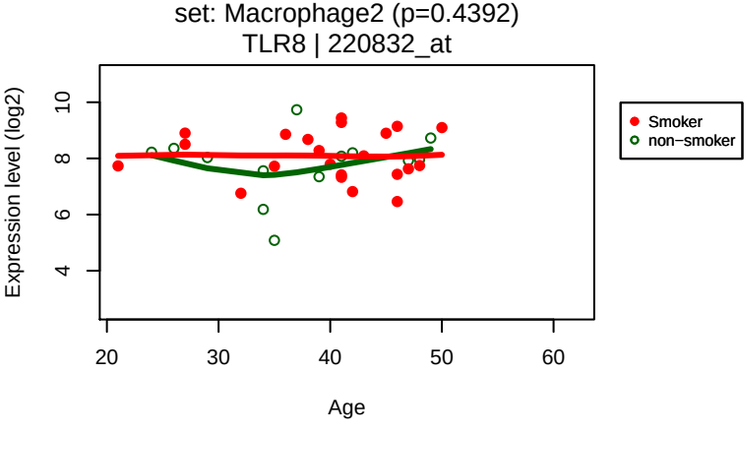

set: SmallAirway2 (p=0.8204)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker

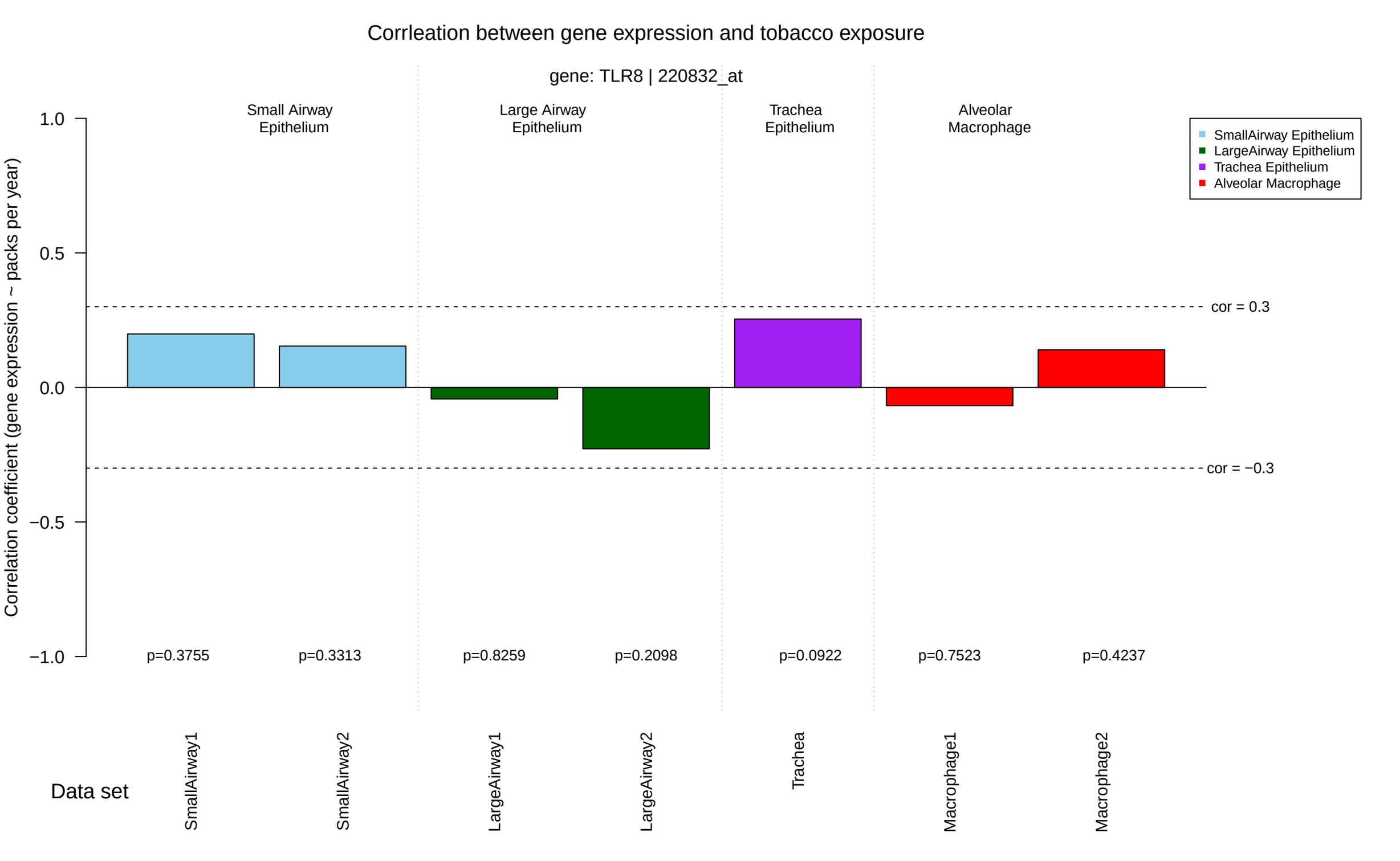

set: LargeAirway1 (p=0.082)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8923)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5314)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.247)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4392)
TLR8 | 220832_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR8 | 220832_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3755
p=0.3313
p=0.8259
p=0.2098
p=0.0922
p=0.7523
p=0.4237
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2